Cybersecurity Lab Environment Setup

Building an isolated virtual lab for penetration testing and ethical hacking practice

Program	Cybersecurity at Networkwalks
Week / Project	Week 01 - Cybersecurity & Pentesting Lab Setup (WK1-PM1)
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Project Overview

This project sets up a virtual cybersecurity and penetration-testing laboratory using VirtualBox and Kali Linux. The purpose of the lab is to create a controlled environment where security tools, network scanning, reconnaissance and vulnerability assessment can be practised safely and repeatedly.

The lab runs on a private virtual network so that additional machines can be added later and used as targets for authorized security testing.

Objectives

- Install and configure VirtualBox.
- Install/import Kali Linux as a virtual machine.
- Create a private NAT Network (10.0.0.0/24) for the lab.
- Configure network connectivity for Kali Linux.
- Assign a consistent IP address (10.0.0.2/24) to the Kali VM.
- Enable clipboard, drag and drop, and a shared Downloads folder.
- Verify network connectivity and DNS resolution.
- Take a clean VM snapshot for recovery.
- Document the complete setup process and prepare for future projects.

Purpose of the Lab

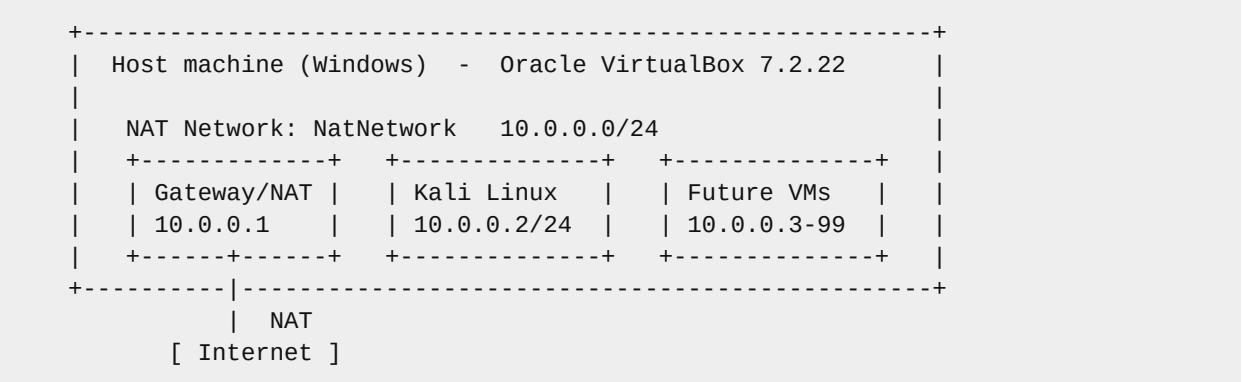
The lab provides an isolated environment for cybersecurity learning and authorized security testing, for example:

- Network reconnaissance and port scanning
- Vulnerability assessment
- Packet analysis
- Web security testing
- Exploitation practice
- Security-tool experimentation

> **Important:** This laboratory must only be used on systems that you own or have explicit permission to test. Do not use the lab or its tools to attack unauthorized systems.

Lab Architecture

Kali Linux (attacker, 10.0.0.2) sits on the VirtualBox NAT Network **NatNetwork** (10.0.0.0/24) together with the gateway 10.0.0.1, which also provides Internet access through the host. Future target VMs join the same network.



Lab Configuration

Component	Configuration
Host OS	Windows
Hypervisor	Oracle VirtualBox 7.2.22
Security OS	Kali Linux (official VirtualBox image)
Kali RAM	4096 MB
Video	VMSVGA, 126 MB video memory, 3D acceleration off
Virtual Network	NAT Network (NatNetwork)
Network Address	10.0.0.0/24 (DHCP enabled, IPv6 disabled)
Kali IP Address	10.0.0.2/24
Default Gateway	10.0.0.1
DNS Server	8.8.8.8
Promiscuous Mode	Allow All
Shared Folder	Host Downloads folder
Future VM Range	10.0.0.3 - 10.0.0.99

Lab Setup Procedure

Step 1. Install 7-Zip

7-Zip was installed to extract the Kali Linux virtual-machine package, which is distributed as a .7z archive.

Step 2. Install VirtualBox

Oracle VirtualBox 7.2.22 (amd64) was installed as the hypervisor. VirtualBox is a Type-2 hypervisor that runs on top of the host operating system and lets several virtual machines share one PC.

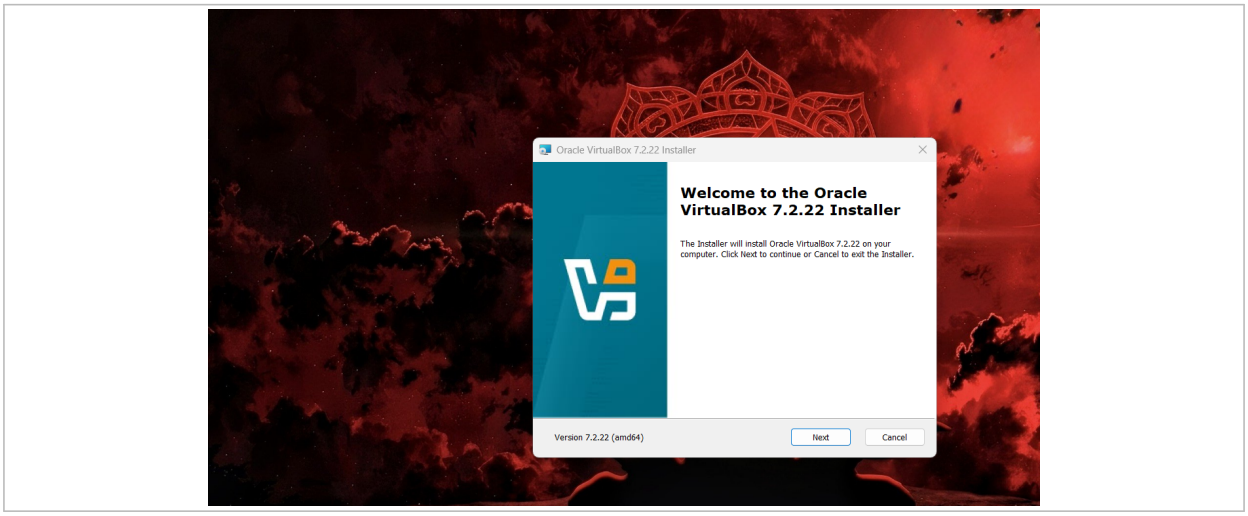


Figure 1: Oracle VirtualBox 7.2.22 installer

Step 3. Create the NAT Network

A dedicated NAT Network was created from **File > Tools > Network** in VirtualBox.

Setting	Value
Network Name	NatNetwork
IPv4 Prefix	10.0.0.0/24
DHCP	Enabled
IPv6	Disabled

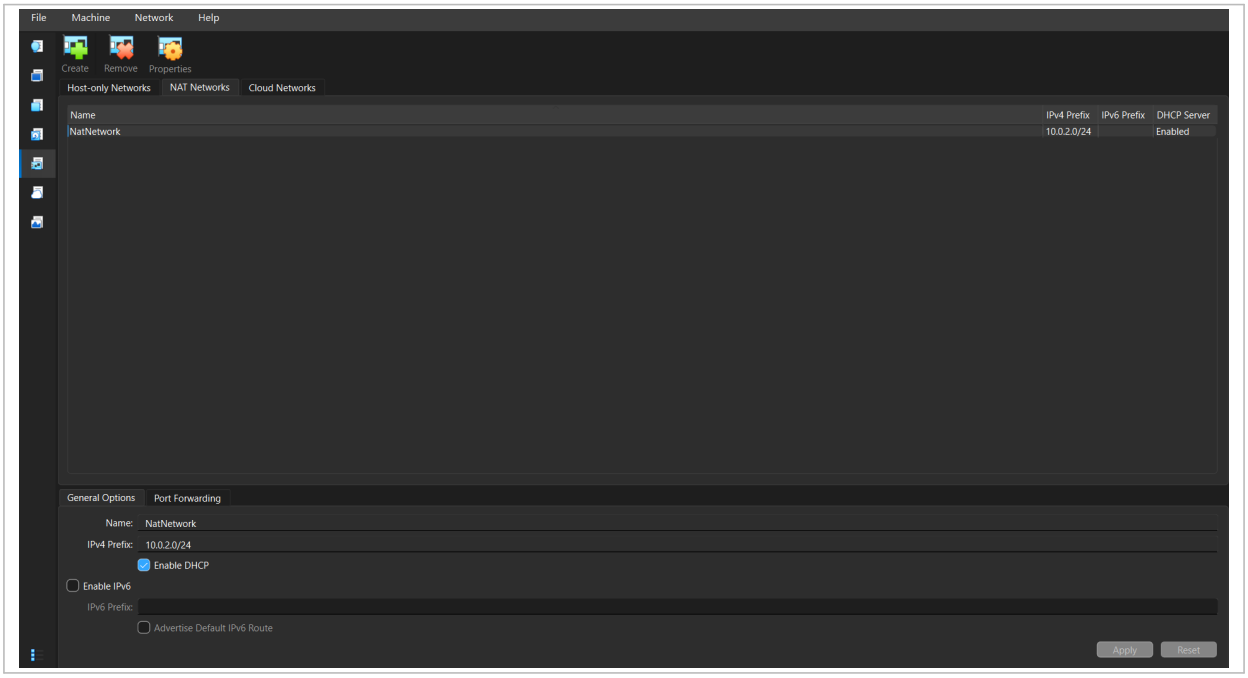


Figure 2: NAT Networks tab in the VirtualBox Network Manager

A NAT Network was chosen because VMs on the same NAT Network can communicate with each other and still have outbound Internet access. This allows future attacker and target VMs to talk inside

the lab.

For comparison, the default **Host-only** network (192.168.56.1/24) connects VMs only to each other and the host, with no Internet access, so it is not used for this lab.

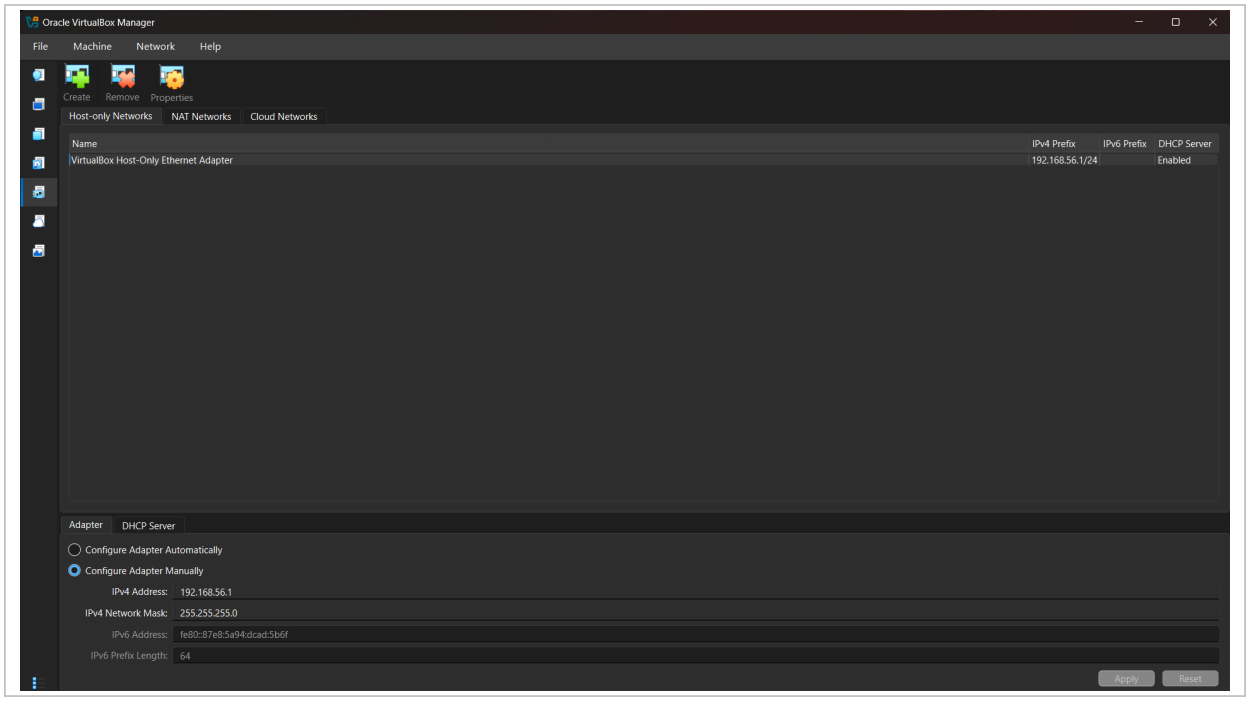


Figure 3: Host-only Networks tab (default adapter 192.168.56.1/24, not used)

> **Note:** VirtualBox creates NatNetwork with the default prefix 10.0.2.0/24. The IPv4 Prefix must be edited to **10.0.0.0/24** and applied so that it matches the Kali static IP.

Step 4. Import and Configure Kali Linux

The Kali Linux VM was downloaded from the official Kali website and imported into VirtualBox. Its network adapter was initially set to Bridged Adapter, which joins the VM to the real LAN through the host's Wi-Fi card:

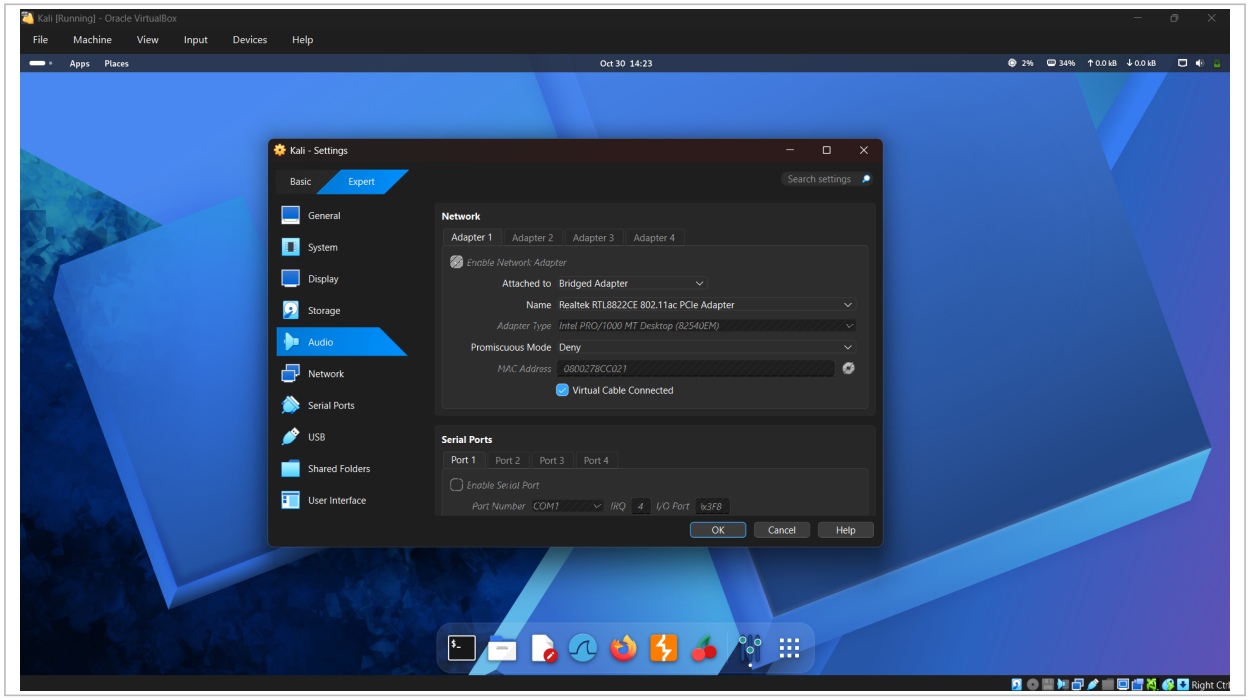


Figure 4: Initial network setting - Bridged Adapter (Promiscuous Mode: Deny)

It was then changed to the lab network:

```
Adapter 1
Attached to:      NAT Network
Name:             NatNetwork
Adapter Type:     Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode: Allow All
```

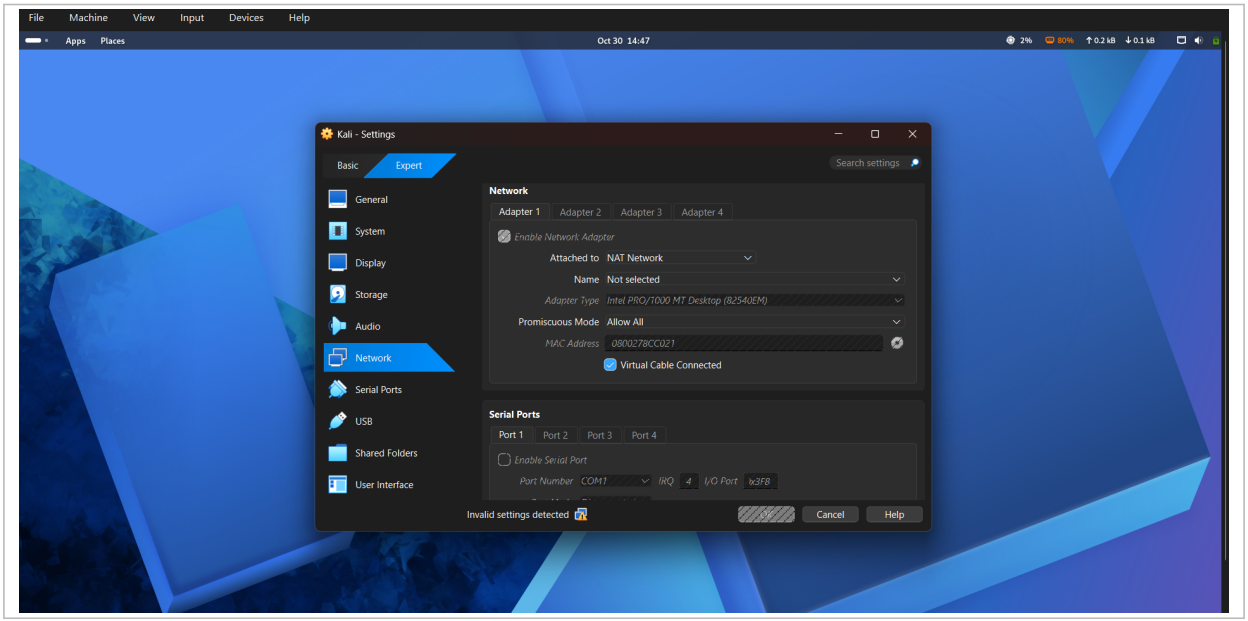


Figure 5: Adapter 1 attached to NAT Network. The Name field must be set to NatNetwork before clicking OK.

Promiscuous Mode: Allow All lets the VM capture traffic addressed to other machines, which packet-analysis tools such as Wireshark need.

System settings

The Motherboard tab shows 4096 MB base memory, chipset PIIX3, no TPM, EFI disabled, I/O APIC and UTC hardware clock enabled. Moving Hard Disk first in the boot order is the usual tidy-up. Settings appear greyed out because the VM was running.

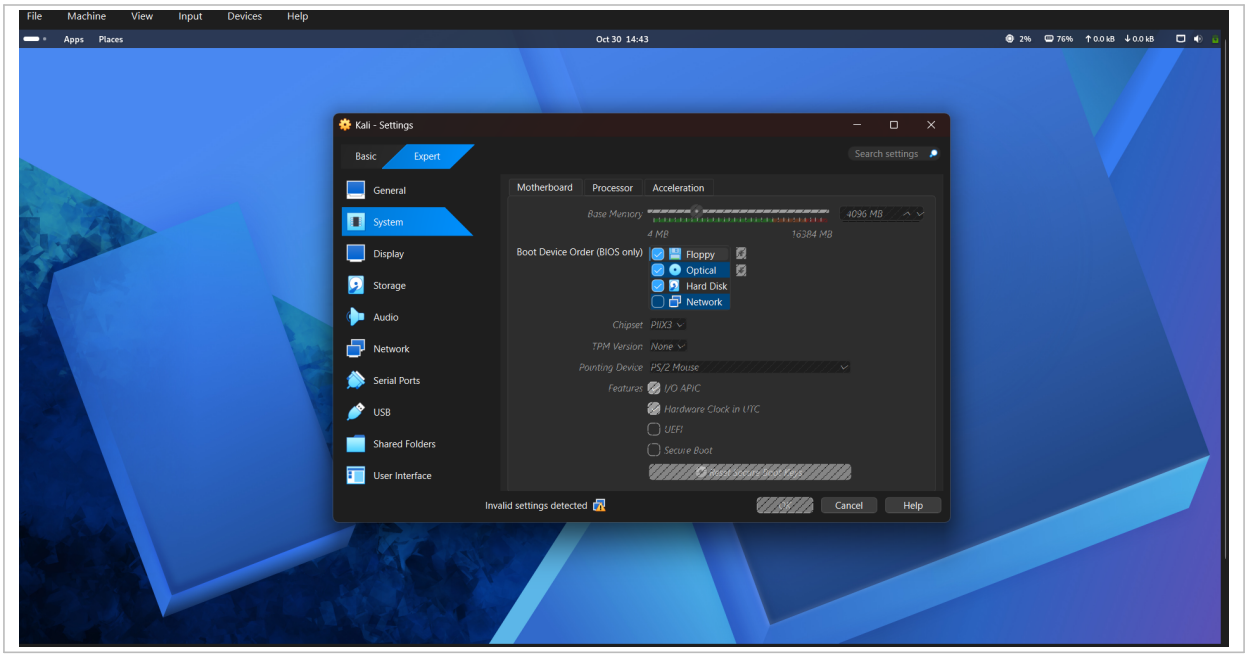


Figure 6: System > Motherboard

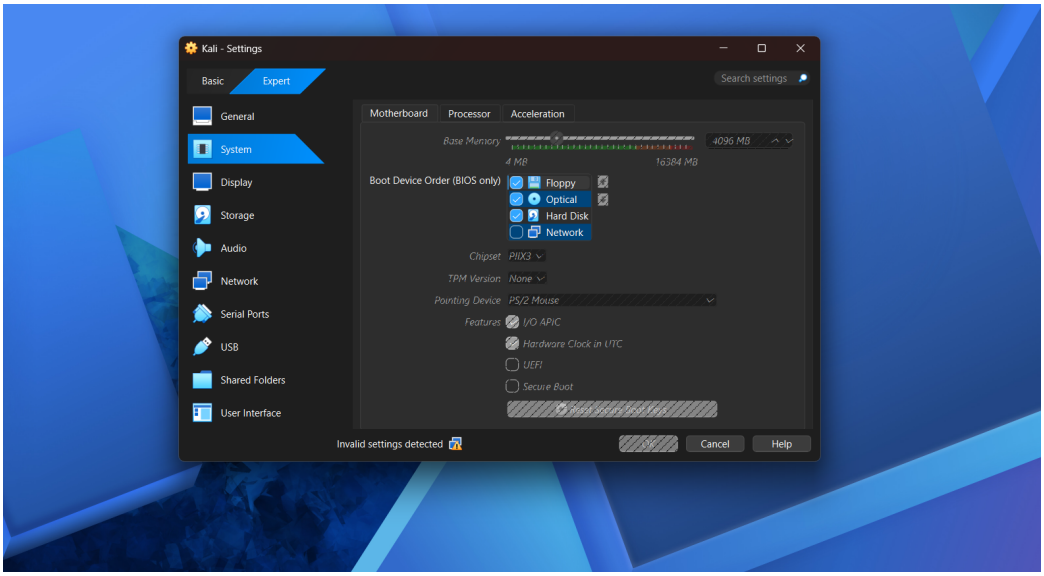


Figure 7: System > Motherboard (cropped view of the same tab)

Nested Paging (hardware virtualization) is enabled on the Acceleration tab. It lets the CPU handle guest memory translation and noticeably speeds up the VM. It depends on VT-x / AMD-V being enabled in the BIOS.

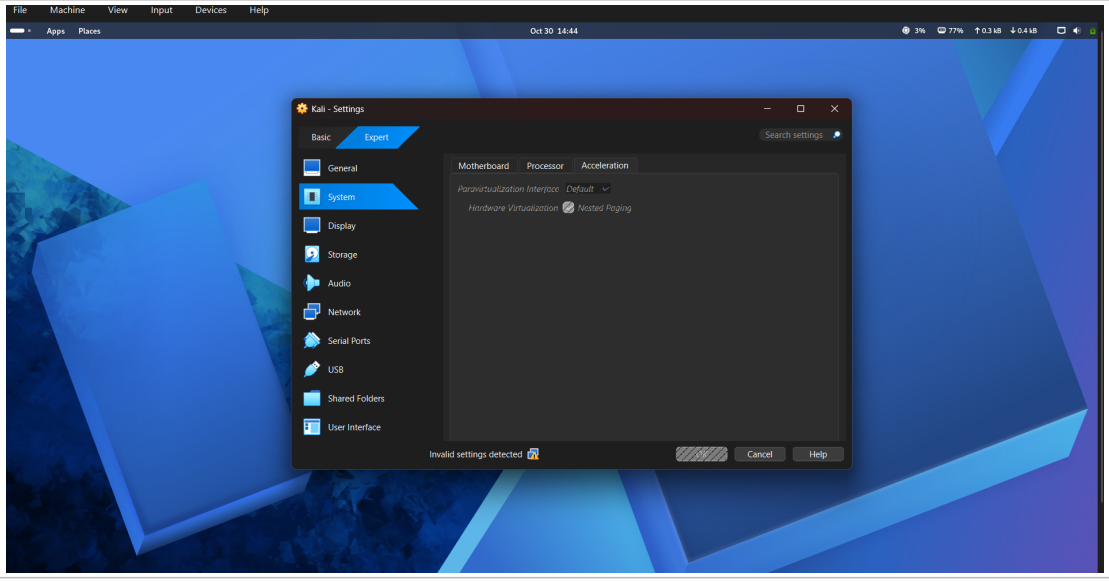


Figure 8: System > Acceleration

In Display, the **VMSVGA** graphics controller is used with 126 MB video memory and 3D acceleration off, which is the stable choice for Kali.

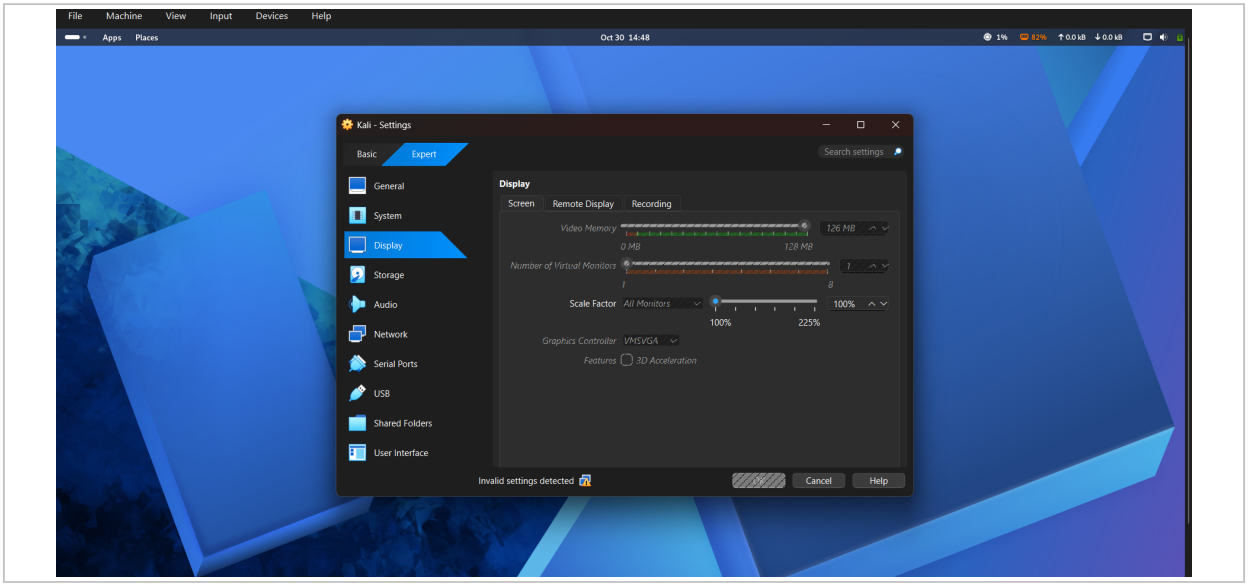


Figure 9: Display settings

Shared folder, clipboard and drag and drop

Under **Shared Folders**, the host Downloads folder is added with Auto-mount and Make Permanent ticked. Inside Kali it appears under `/media/sf_Downloads`; add the user to the `vboxsf` group (`sudo usermod -aG vboxsf $USER`) and reboot. Clipboard and drag and drop are set to **Bidirectional** in **General > Advanced**.

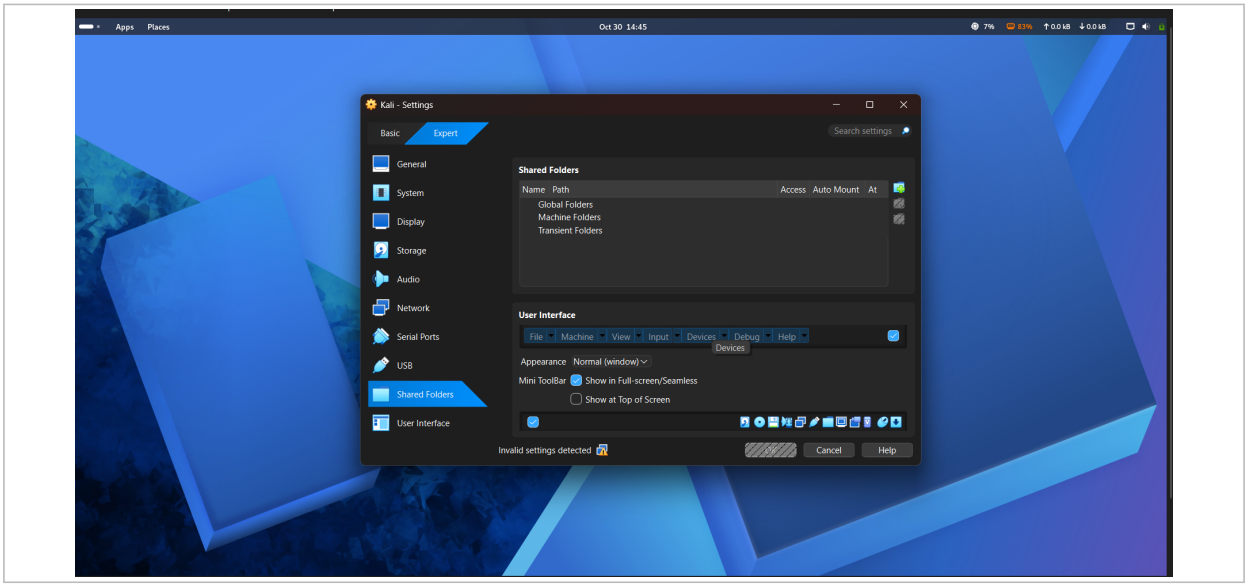


Figure 10: Shared Folders and User Interface settings

Step 5. Configure the Kali Linux Network

Kali was given a fixed IPv4 address so that it is easy to document and reference in later exercises.

```
IP Address:  10.0.0.2
Subnet Mask: 255.255.255.0  (/24)
Gateway:     10.0.0.1
DNS:         8.8.8.8
```

This can be done in the GUI (network icon > Edit Connections > IPv4 Settings > Manual) or from the terminal:

```
sudo nmcli connection modify "Wired connection 1" \
    ipv4.method manual ipv4.addresses 10.0.0.2/24 \
    ipv4.gateway 10.0.0.1 ipv4.dns 8.8.8.8
sudo nmcli connection down "Wired connection 1"
sudo nmcli connection up "Wired connection 1"
```

Step 6. Create a Clean VM Snapshot

After the initial configuration worked, a VirtualBox snapshot was taken (example name: *Clean Kali - Network Setup*). It is the clean baseline of the lab: if a later exercise damages the VM, it can be restored to this state.

Lab Verification

Test	Command	Expected Result
Check IP address	ip a	10.0.0.2/24 shown on eth0
Test gateway	ping -c 3 10.0.0.1	Successful replies
Test Internet	ping -c 3 8.8.8.8	Successful replies
Test DNS	nslookup networkwalks.com	Domain resolves
Verify Nmap	nmap --version	Nmap version displayed
Verify snapshot	Restore snapshot, run ip a	Baseline configuration restored

Problems Encountered & Solutions

Problem 1. No Internet After Network Start-up

On VirtualBox 7 with recent Kali versions the connection can stall at start-up because NetworkManager performs Duplicate Address Detection (DAD) on the IPv4 address. The fix is to disable the DAD timeout and restart the connection:

```
sudo nmcli connection modify "Wired connection 1" ipv4.dad-timeout 0
sudo nmcli connection down "Wired connection 1"
sudo nmcli connection up "Wired connection 1"
```



Figure 11: nmcli fix on Kali. Earlier attempts failed with 'unknown connection' because the spaces in the name were not escaped; the final command reported 'Connection successfully activated'.

> **Important:** Connection names differ between systems. Check yours with `nmcli connection show` before running the commands, and quote the name or escape spaces as `Wired\ connection\ 1`.

Problem 2. Settings Dialog Shows 'Invalid settings detected'

After switching Adapter 1 to NAT Network, the Name field showed *Not selected* and the OK button stayed disabled. Selecting **NatNetwork** in the Name list (the network must exist under Tools > Network) resolves it.

If the VM fails to start at all with a VT-x / AMD-V error, hardware virtualization is disabled in the BIOS/UEFI: restart, enable Intel VT-x (or AMD-V / SVM), save, and boot again.

What I Learned

1. NAT vs NAT Network

Plain NAT isolates each VM from the others. A NAT Network puts VMs on one shared private subnet where they can talk to each other and still reach the Internet, which is what a multi-machine lab needs.

2. Virtual Machine Networking

Bridged, Host-only and NAT Network modes connect a VM differently, and the choice decides who the VM can reach and who can reach it.

3. Static IP Configuration

I learned to configure and verify the IPv4 address, subnet mask, gateway and DNS in Kali Linux, and to use ping tests to tell routing problems from DNS problems.

4. VM Snapshots

A clean snapshot taken before risky or experimental work gives a known-good recovery point.

5. Documentation

Recording commands, settings, screenshots, problems and fixes is a core part of a professional cybersecurity project.

Security & Ethical Use

This laboratory is intended strictly for educational purposes and for systems that you own or are authorized to test.

Tools & Resources

- 7-Zip: https://7-zip.org/download.html
- VirtualBox: https://virtualbox.org/wiki/Downloads
- Kali Linux: https://kali.org/get-kali

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